

Saidakhmad Nuriddinov

Autonomous Driving Simulation Engineer

CONTACT

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SKILLS

Robotics & Simulation

NVIDIA Isaac Sim, Omniverse, USD, PhysX, OmniGraph, ROS 2, Nucleus

Languages

Python, C++, TypeScript, JavaScript

Frontend

React, Next.js, SCSS

Backend

NestJS, Node.js, Flask

Database & DevOps

PostgreSQL, MySQL, MariaDB, Docker, Vercel, GitHub Pages

Tools

Git, Postman, Notion, Logitech G29

LANGUAGES

Korean Advanced · TOPIK 4
English Advanced · IELTS 7
Russian Intermediate

CERTIFICATES

KIIP Level 5
Nov 2025
Wcoding Full-Stack Bootcamp
Nov 2024
TOPIK Level 4
Jul 2023
Business Korean, Seoul Global Center
Sep 2023
Minister's Prize, Education Ministry of Uganda
Jun 2023
President's Award, KIU Uganda
Jun 2023
Dean's Award, Computer Engineering
Dec 2021

EDUCATION

Gachon University

Seongnam, South Korea · Aug 2018 – Feb 2023

B.S. in Computer Engineering · GPA 3.54 / 4.5

EXPERIENCE

Maum.ai (마음에이아이)

Seongnam · Feb 2026 – Present

Autonomous Driving Simulation Engineer · WoRV

Autonomous yard-tractor simulation (`ability_port`, XCMG 15-DOF) on NVIDIA Isaac Sim / Omniverse.

- ▶ Built **worv.env.lights**, a solar lighting Isaac Sim extension using a stdlib-only Fourier model; established the MVC pattern (extension → window → controller) adopted as the team-wide template for all `worv.*` extensions.
- ▶ Shipped **worv.env.climate** (rain/snow/fog via USD `PointInstancer` + Warp + RTX) and led a 5-bug fix branch introducing a 3-source **PositionTracker** for robot-following rain and controller-level error surfacing.
- ▶ Designed **worv.core.logging**, a configurable-Hz simulation logger combining OmniGraph with direct PhysX USD reads via a pluggable **BaseExporter** — enabled the team's first benchmark scenario comparison.
- ▶ Built a **Docker Nucleus worker** plus a 3-commit **UE5 C++ plugin patch** fixing UE Connector's baked 1×1 opacity textures so designers can edit opacity directly in Isaac Sim.
- ▶ Applied physics, colliders, and NavMesh across the `ability_port` map; added an Ackermann controller and refactored the ROS 2 `teleop_node` from Xbox-only to hardware-agnostic via a **JoyMapping** dataclass.

Stack: NVIDIA Isaac Sim · Omniverse · USD · PhysX · OmniGraph · Nucleus · ROS 2 · Python · C++ (UE5) · Docker

EARLIER EXPERIENCE

ChoiceTech Korea (초이스테크코리아)

Yongin · Jul 2025 – Oct 2025

Backend Developer

AI-powered hair & skin analysis platforms used by L'Oréal and Dior.

- ▶ Implemented JWT auth, CRUD, and pagination on **NestJS + PostgreSQL**; shipped new features and bug fixes that improved backend stability and maintainability.
- ▶ Customized backend APIs and database structures for **L'Oréal and Dior** analysis platforms; implemented real-time update mechanisms tailored to each brand.
- ▶ Validated web/app analysis-result consistency on **Next.js**, integrated REST APIs, and resolved cross-stack data-flow issues end-to-end.
- ▶ Standardized license-expiry logic across all projects with automated daily updates and 30-day pre-expiration notifications.
- ▶ Optimized SQL keyword mapping for dermatological analysis, achieving **over 20% faster** real-time response.

Stack: NestJS · PostgreSQL · Next.js · React · TypeScript · JavaScript

STEMON (스텨몬)

Seongnam · Dec 2024 – Jul 2025

Software Developer

Web services and embedded medical-device management systems.

- ▶ Built a corporate **Q&A system** (HTML/CSS/JS) and an investor-relations **React/SCSS** SPA; deployed via GitHub Pages and Vercel with CI/CD.
- ▶ Developed the **DDDS Admin App** in C++ for real-time medical-device monitoring with auto-detection of faulty components and CSV export.
- ▶ Validated hardware-software integration with **ESP32-S3** over USB serial; redesigned the on-device UI on DWIN DGUS for production deployments.
- ▶ Led the final-month **team project** as Wcoding bootcamp lead — full-stack React + Flask + MySQL with auth, notifications, and DB design.

Stack: React · SCSS · JavaScript · HTML/CSS · Flask · MySQL · C++

SELECTED PROJECTS

worv.env.lights Solar lighting Isaac Sim extension. Spencer 1971 Fourier solar model (stdlib only), Tanner Helland Kelvin → RGB, timeline 1×–500×, adopt-vs-create lights, stage lifecycle subscription. MVC architecture became the team-wide template.	Maum.ai · 2026.03	worv.env.climate Rain/snow/fog effects via USD PointInstancer + Warp kernels + RTX carb settings. 5-bug fix branch (+405/–358) including PositionTracker, RTX key discovery, silent-failure surfacing pattern.	Maum.ai · 2026.04
worv.core.logging Configurable-Hz simulation data logger. Hybrid OmniGraph + direct PhysX USD reads. Pluggable BaseExporter (CSV/JSON/Console), smart root-link resolution, frame dedup via <code>absoluteSimTime</code> .	Maum.ai · 2026.04	usd-opacity-tools Pipeline Docker Nucleus post-processor + 3-commit UE5 C++ plugin patch fixing UE Connector's baked 1×1 opacity textures so designers can edit opacity in Isaac Sim. Includes DeviceFlow auth workaround for headless containers.	Maum.ai · 2026.04
G29 Teleop Port Hardware-agnostic ROS 2 teleop for the XCMG tractor. Refactored hardcoded Xbox enum into a configurable JoyMapping dataclass with per-axis inversion, deadzone gating, accel/brake pedal merging.	Maum.ai · 2026.03 – 2026.04	DDDS Admin App Internal real-time device monitoring (C++) with ESP32-S3 USB serial integration and DWIN DGUS embedded UI; auto-detection of faulty parts and CSV data export.	STEMON · 2025.03 – 2025.07

EXTRACURRICULAR

Full-Stack Bootcamp, Wcoding Team Lead · Jul – Nov 2024	Business Korean Program Seoul Global Center · Jul – Sep 2023
Leaders Conference, IYF Team Leader (Uganda) · 2023	Teaching Assistant Gachon University · Mar – Jun 2022